

Section 7.1: Communications Applications

One of the positives of ICT is that it allows us to communicate with each other in many different ways.

This section looks at some of these ways in which we communicate using ICT and focuses on:

- 1 Newsletters 2 Websites 3 Multimedia Presentations**
- 4 Music Scores 5 Cartoons 6 Flyers**
- 7 Posters**

Key Concepts of this section:

- # Understand the two different types of communication - **Paper-Based** and **Digital**.
- # Know the various **methods of communication** using both paper-based and digital mediums.
- # Be able to discuss how **ICT** has **changed how we communicate information**.

Communicating using ICT

Different types of communication

- # There are 2 different types of communication where ICT can help us share information or ideas:

- 1. Personal Communication**
- 2. Business Communication**

Personal Communication Examples:

- Telling a friend when and where to meet up
- Reminding your brother about a family member's birthday

Business Communication Examples:

- Letting customers know about a sale
- Advertising a new product
- Sending messages to employees or colleagues



Key Words:

Personal Communication,
Business Communication



ICT can be used to **communicate** different types of **personal information**

Paper-Based Communication

What is paper-based communication?

- # This type of communication involves **producing hard copy's (printouts) of documents**.
- # Creating documents suitable for **printing** is often called **Desktop Publishing (DTP)**.
- # Advantages/Disadvantages of paper-based communication are summarised below:

Advantages

- Document can be brought to people who are disabled (e.g. newspaper).
- Document can be printed in Braille for blind people.
- The document can be carried around and read whenever you want.

Disadvantages

- Costs time and money to distribute the document.
- It is not possible to add exciting effects to a printed document (animation, video etc).
- Can be very expensive to print off copies of the document.

Examples of paper-based communication

- # Examples of paper-based communication are listed below:
 - **Newsletters**
 - **Posters**
 - **Flyers**
 - **Newspapers**
 - **Magazines**
 - **and lots more.....**

1 Newsletters

- # Newsletters are **released regularly** (e.g. once a month) and are used to **inform people about areas of interest** to them.
- # Schools often release newsletters to keep parents **up-to-date with school events** and important **dates** (like examinations).
- # Newsletters can be created very easily using DTP software and can be printed using a standard home printer.



Key Words:

Printouts, Newsletters,
Posters, Flyers,



Printing off **hard-copies of documents** can be very **expensive** especially if they are colour



A school newsletter detailing future events

2 Posters

- # Posters are a simple but effective way of communicating with people, for example:
 - **Promoting an event (like concerts or festivals)**
 - **Promoting good causes (like charity sales etc)**
 - **Advertising business promotions or sales**
 - **Decoration which show people your interest (e.g. bedroom posters of movie stars)**
- # Posters include combinations of **Text, Images, Colours** and come in all **different sizes**.
- # Very **large posters** are **printed onto multiple sheets of paper** and then joined together (like on a billboard)
- # Posters are usually **attached onto a wall** in easy to see positions.



A large poster advertising Coca Cola

3 Flyers

- # Flyers are very **similar to posters** except they are **smaller in size**.
- # Flyers are normally printed onto **single sheets of paper, 8 1/2" x 11"** in size.
- # They are usually **handed out to people** instead of being attached to a wall. The small flyer can then be easily put into your pocket and read later.



Flyers are '**small posters**' that can be handed out

Digital Communication

What is digital communication?

- # Digital communication **requires a computer and a screen** in order to present the document.
- # Digital communication involves creating documents and files and then '**displaying**' them via a computer and a screen.
- # Advantages/Disadvantages of digital-based communication are summarised below:

Advantages

- Far cheaper as there is no printing.
- Exciting effects can be added (animation, sound video etc).
- Easier to update content.
- Communication can be interactive (viewer can click options or leave comments etc)

Disadvantages

- Often requires expensive software.
- Some software is very difficult to use (music production, website design etc).
- More complex documents require more time to build.

Examples of digital communication

- # Examples of digital communication are listed below:
 - **Websites**
 - **Multimedia Presentations**
 - **Music Scores**
 - **Cartoons**



Key Words:

Computer Hardware /
Software, Website, Multimedia,
Music Score, Cartoon /
Animation



Digital communication gives the user an **interactive experience** with **sound, video** and **click able areas**

1 Websites

Websites are made up of a **collection of web-pages** and can contain a wide variety of content. For example:

- **Text**
- **Images**
- **Sound**
- **Animation**
- **Video**
- **Clickable areas like hyperlinks**
- **User Interactivity (comment boxes, forums etc)**

Many companies choose to **advertise their products on websites** rather than on paper-based documents (like posters). This is because websites can be **easily accessed by millions and millions of people**.

Imagine the costs of printing millions and millions of posters!!

Modern websites include lots of **multimedia content** like **streaming video, sound** and **animation**. This makes communicating information over websites a much more **interesting experience**.

Advantages/Disadvantages of communicating over websites are summarised below:

Advantages

- Multimedia content can be added (sound, video etc).
- Web Pages can be created or updated and then published at no extra cost (no printing for example).
- The content is accessible to literally millions of people at any time.

Disadvantages

- You need a computer or portable device to view the website (expensive).
- You need an Internet connection in order to view the website. (Internet connections can go down)
- You need specialist training in order to build the website.
- Websites can be hacked or become infected with viruses.



NASA online communicates news to us via **multimedia content** like **videos, images, text** and **sound**
(Click image to visit the website)

NOTE!

When the world wide web was invented in 1991, early websites did not include much multimedia at all. They were only capable of basic text and images.

The image below shows you the first web page ever created:



(Click the image to visit the world's first web page)

2 Multimedia Presentations

'Multimedia' is the term given to documents/software that contain combinations of:

- **Text**
- **Images**
- **Animations**
- **Sound/Music**
- **Video**

- # Documents that make use of multimedia are **much more interesting than paper-based documents** that cannot make use of multimedia.
- # Multimedia presentations are often used in education because they '**bring subjects to life**' which helps to **make learning fun**.

Old printed text books that cannot use multimedia can become very dull and boring if you study from them for too long!

- # Advantages/Disadvantages of communicating over multimedia presentations are summarised below:

Advantages

- Sound, animations and video can be used.
- Make topics more interesting and helps hold the user's attention.

Disadvantages

- You need a computer to view the presentation (expensive).
- Software needed to create the presentation can be expensive.
- Specialist training may be needed in order to build the presentation.

3 Music Scores

- # Music can now be **composed and edited on a computer** using special hardware/software.
- # Methods include the following:

1. **Sequencers** → Used to create **musical sounds** like **drum beats**.
2. **Sound Wave Editors** → Allow **music/sound to be edited and modified** (e.g. raise tempo, add effects etc)
3. **MIDI** → MIDI (Musical Instrument Digital Interface) allows **musical instruments to be plugged into a computer**.

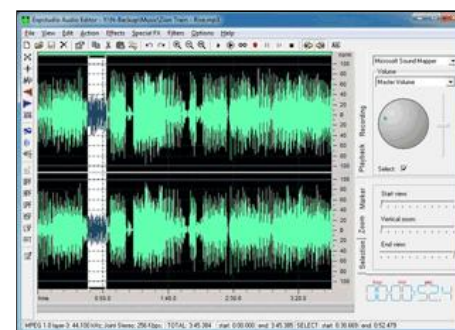
Analogue data from the instrument is **converted into digital data** by an **ADC**.

Once in the computer, the digital music can be **edited and stored**.

The computer can re-send the digital music back to the



Interactive encyclopedia's use **multimedia** to **bring topics to life** and **make learning fun**



Music editors allow us to **enhance sounds** with different **effects** or **edit tempo, pitch** etc

instrument using DAC to convert to analogue. The instrument (keyboard for example) could then play the edited music back.

4. Digital Music Notators

➔ This software allows us to **compose music on a computer** rather than writing notes by hand. Notes can be entered by:

- **Typing them into the software**
- **Input via MIDI**
- **Scanned in from existing paper-based notes**



Music does not need to be written by hand anymore



Digital Music Notators allow music to be composed on a computer



A guitar connected to a laptop via **MIDI**. This allows notes created by the guitar to be **recorded and edited**

4 Cartoon Animations

- # Cartoon development has changed a lot with the introduction of ICT.
- # **Old cartoons** used to have to be **drawn by hand frame-by-frame**. Every tiny movement of a character or object had to be drawn. Lighting and fine detail was also added by hand. This was a **hugely time consuming process**.
- # **Computers** can hugely **speed up this process** by using **3-D models** which are made up of **mathematical polygons**.
- # Special software can be used to add the following to the 3-D model:
 - **Textures** - adding skin, clothes etc
 - **Poses** - positioning the model into different shapes across frames
 - **Effects** - adding lighting, shadows etc
 - **Animation** - combining each frame to simulate movement.



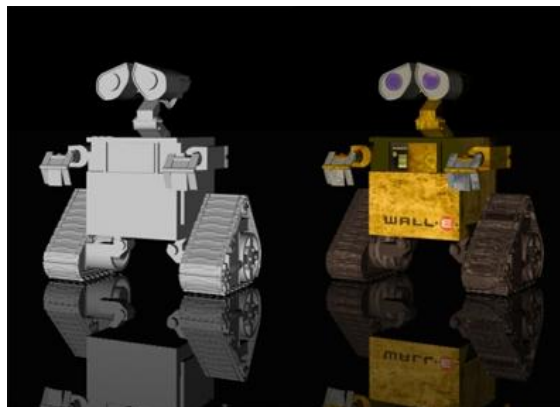
An original **hand drawn cartoon frame** from Disney's Snow White and the Seven Dwarfs

NOTE!

You can use simple animation software to



3-D model created on a computer



Pixar's Wall-E 3-D model with and without textures

create your own cartoons on your computer at home.

'Flash' is a popular option and it allows you to use **Key Frames** and a technique called **Tweening** to make objects move.

[Click for Picture Example](#)

Interactive Communication

Allowing users to respond to information

- # Interactive communication is where people can respond to the information that they are given. Computers have made this much easier than ever before.
- # Modern '**Web 2.0**' websites such as **blogs**, **wikis** and **social networking sites** offer users the ability to interact in the following ways:
 - **Leave comments**
 - **Thumbs Up / Thumbs Down** - indicate if they liked the information or not
 - **Ratings** - score out of 5 for the usefulness of information for example
 - **Alter the content** - wikis allow users to actually change the information

NOTE!

Blogs, Wikis and Social Networking Sites were covered in more detail in [Section 6 - The Effects of ICT](#)



You Tube uses **thumbs** to indicate what **viewers** thought of uploaded videos.



Key Words:

Interactive, Web 2.0, Respond, Blog, Wiki, Social Networking



Interactive '**Web 2.0**' websites allow **users** to **respond to information** that they receive



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